

SUMMARY

Machine Learning Engineer & AI Researcher with expertise in building scalable, production-ready AI systems and multimodal deep learning pipelines. Proven track record of engineering ETL pipelines and MLOps workflows, optimizing model inference latency, and deploying RESTful microservices using PyTorch, FastAPI, and cloud platforms (Azure/AWS). Adept at translating complex geospatial and logistical data into high-impact, quantifiable solutions, with a strong foundation in computer vision, NLP, distributed training, and end-to-end model lifecycle management.

EDUCATION

Indian Institute of Technology Kharagpur	B.S. in Exploration Geophysics	2024 – 2028
CGPA: 8.30/10		
New Holy Ganges Public School, Bihar	CBSE Class XII	2024
Percentage: 89%		
Meluha International School, Telangana	CBSE Class X	2022
Percentage: 97%		

EXPERIENCE

•Indiana University Bloomington	May 2025 – Present
AI Research Intern – Natural Disaster Monitoring	Remote
– Engineered a multimodal fusion pipeline combining RGB/thermal imagery and LiDAR point clouds to detect flood extents and tornado structural damage, reducing manual labeling effort by 40% compared to single-sensor baselines using distributed training across multi-GPU nodes	
– Fine-tuned a LLaVA-style vision-language model on 3 disaster scenario domains (flood, wildfire, tornado), achieving zero-shot scene classification across unseen geographic regions with over 80% accuracy via cross-functional collaboration with faculty researchers	
– Built automated ETL pipelines and remote sensing analysis workflows in Python to process satellite and drone imagery at scale, cutting per-image analysis time from 8 minutes to under 30 seconds with scalable architecture supporting high-throughput batch inference	
– Designed benchmark evaluation protocols and model monitoring dashboards for multimodal disaster AI systems, producing reusable evaluation scripts adopted across 2 concurrent lab projects following Agile sprint cycles	

PROJECTS

•FADS – First-Attempt Delivery Score Prediction	March 2026 – April 2026
ML MLOps ETL Pipelines LightGBM XGBoost FastAPI Docker MLflow	GitHub
– Designed an end-to-end ML pipeline with ETL data engineering on Olist and LaDe last-mile logistics datasets, engineering geo, weather, and traffic features into a structured SQLite feature store, cutting feature retrieval latency by 3x versus raw CSV processing with a scalable microservices architecture	
– Trained and auto-selected the best model across LightGBM, XGBoost, and Random Forest tracked via MLflow with full experiment reproducibility, achieving a FADS prediction AUC of 0.87 on held-out delivery records; integrated data drift detection alerts for production model monitoring	
– Architected a 5-tier Human-in-the-Loop dispatch system (AUTO_DISPATCH to FULL_HOLD) using predicted risk scores, enabling automatic clearance of 80%+ of orders and routing only high-risk deliveries to human review queues within a cross-functional workflow	
– Deployed the full system as a FastAPI REST microservice + Streamlit dashboard + WebSocket live stream via Docker Compose, supporting real-time scoring of 1,000+ delivery events per minute in a local load test with high-throughput distributed processing	
•Seismic Fault Detection using U-Net	Feb 2026 – March 2026
Deep Learning PyTorch Computer Vision Distributed Training Geophysics	GitHub
– Built a U-Net segmentation architecture in PyTorch for 9-class dense pixel labeling (7 stratigraphic layers + fault surfaces) on 3D seismic volumes containing over 10 million labeled voxels , leveraging scalable distributed training techniques	
– Engineered a Hybrid Loss function (Weighted Cross-Entropy + Dice Loss) to handle extreme class imbalance, where fault pixels constitute less than 0.01% of the dataset , improving fault recall by 34% over vanilla cross-entropy with automated data drift monitoring on validation splits	
– Designed a memory-efficient sliding-window inference pipeline for full 3D volume reconstruction on a single 8GB GPU, achieving a Mean Dice Score of 87.8% across all 9 classes without out-of-memory errors; packaged into a reproducible, production-ready RESTful API microservice	
– Managed 350MB+ model checkpoints via Git LFS with documented training, inference, and evaluation scripts following Agile sprint-based development for clean CI/CD integration	
•Kavach AI – Voice Fraud & Deepfake Detection System	Jan 2026 – Feb 2026
AI/ML HuBERT Whisper FastAPI Azure CI/CD	GitHub

- Built a **3-model fusion inference pipeline** combining HuBERT acoustic embeddings, Whisper ASR transcripts, and Sentence Transformer semantic vectors to score voice fraud risk across **2 distinct attack classes** (deepfake and social engineering) with real-time model monitoring
- Optimized the **scalable inference stack** to generate fraud risk scores and full audit reports in **under 10 seconds** per audio clip, making real-time call-center deployment feasible at production high-throughput
- Designed **RESTful API development in FastAPI** with a microservices architecture and a React-based analyst dashboard displaying risk scores, transcripts, and waveform visualizations, reducing manual review time per flagged call by **60%**
- Automated build, test, and deployment via **GitHub Actions CI/CD** following Agile sprint methodology, shipping to Microsoft Azure App Service with zero-downtime rolling updates across staging and production environments

•**Smart CloudOps Platform**

Feb 2026 – March 2026

Full Stack / Cloud Engineering / AWS / Azure / GCP / Anomaly Detection / Scalable Architecture

Github

- Built a **multi-cloud cost aggregation system** ingesting AWS, Azure, and GCP billing APIs into a **unified normalized schema** via ETL data engineering pipelines, enabling cross-provider spend comparison across **3 cloud environments** from a single dashboard with scalable distributed architecture
- Designed **real-time analytics pipelines** processing hourly usage events, surfacing the **top 5 cost drivers per service** and flagging resource idle patterns accounting for **22% of simulated monthly spend**
- Implemented a **statistical anomaly detection and model monitoring module** using rolling z-score thresholds to alert on abnormal spending spikes within **1 hour of occurrence**, versus 24-hour delays in native cloud billing consoles

TECHNICAL SKILLS

- **Programming Languages:** C | C++ | Python
- **Libraries / Frameworks:** PyTorch | Scikit-learn | LightGBM | OpenCV | NumPy | Pandas | FastAPI | Flask | Streamlit
- **MLOps & Data Engineering:** MLflow | Docker | GitHub Actions CI/CD | ETL Pipelines
- **Tools:** Git | Figma

COURSEWORK

- **CS & Machine Learning:** Programming and Data Structures*, Essence of Machine Learning
- **Geoscience:** Structural Geology, Geophysical Signal Processing, Mineralogy, Geophysical Field Theory, Seismology
- **Mathematics:** Linear Algebra, Advanced Calculus, Probability and Statistics
- **MOOCs:** Data Structures and Algorithms, Machine Learning and Deep Learning, Generative AI

POSITIONS OF RESPONSIBILITY

•**Student Member, Students’ Branding and Relations Cell,**

Aug 2024 – Jan 2025

– Led cross-functional outreach for the **Young Innovators Program (YIP)**, contacting over **15 schools** across Bihar and Telangana to drive student registrations, contributing to a **20% increase** in participation versus the prior cohort

– Designed promotional materials and coordinated digital campaigns that increased **social media reach** for the YIP launch event by approximately **30%**

•**Wing Representative, Jagdish Chandra Bose Hall of Residence,**

Sept 2024 – Present

– Served as the primary liaison for **40+ residents**, resolving maintenance, academic, and administrative grievances with hall management within a **48-hour turnaround** on average

– Organized **3 wing-level events** per semester including study groups and sports sessions, raising wing participation in hall activities by approximately **25%**

ACHIEVEMENTS & EXTRA-CURRICULAR

- Won **Bronze Medal** at the Open IIT Kharagpur Table Tennis Championship, competing against **1800+ participants**
- Cleared multiple competitive rounds in **GenAI Hackathon** (Economic Times), **AI Unlocked** (Microsoft), and **Zomatathon**, placing in the top shortlists out of hundreds of national applicants
- Served as **NCC Cadet**, completed residential NCC Camp, and secured **Silver Medal** in inter-unit shooting competition